

Abdullah Khaled

Engineer-in-training focused on autonomy, vision, and embedded systems

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Education

Wakeland High School, High School Diploma – Frisco, TX Aug 2023 – May 2027

- Unweighted GPA: 4.0
- 18 AP courses by graduation (completed and planned)

Frisco ISD Career and Technical Education Center, Project Lead The Way – Frisco, TX Aug 2023 – May 2027

- Frisco ISD Career and Technical Education Center Project Lead the Way (PLTW) Nationally Distinguished Engineering Program
- Courses: PLTW Introduction to Engineering Design (9th grade), PLTW Principles of Engineering (10th grade), PLTW Digital Electronics (11th grade), PLTW Engineering Design and Development (12th grade)
- Engineering Portfolio: [akhaled247.github.io/pltw-engineering](https://github.com/akhaled247/pltw-engineering)

Engineering Projects

OralVision (Oral Disease Detector) – Conrad Challenge; Diamond Challenge Oct 2025 – present

- Developed privacy-preserving oral disease screening pipeline (centrally trained model + on-device inference, without transmitting raw patient images), with my engineering focus on the imaging device end-to-end while my partner focused on the inference/training layer
- Designed, wired, and programmed the imaging device (XIAO-ESP32-C6 MCU + OV5642 5MP camera), including power-aware capture (camera current draw can trip the 5V voltage regulator)
- Bench-tested image capture quality and robustness across lighting/viewing angles to maximize reliable low-cost imaging
- Verified repeatability by running multiple capture sessions under comparable conditions and confirming output consistency
- Iterated through 3 PCB revisions and 5 3D-printed snap-fit case revisions to improve reliability and manufacturability
- Used Onshape 3D CAD software to design and 3D print the device, reducing manufacturing costs
- Conrad Challenge Innovator Award and Alternate Finalist
- Diamond Challenge Semi-Finalist
- Patent application in progress with United States Patent and Trademark Office (USPTO)

WatchFall (Senior Fall Monitor Device) – Samsung Solve for Tomorrow Sept 2025 – present

- Designed fall-detection device using Raspberry Pi Pico with on-device TensorFlow Lite inference
- Developed lightweight ML model for real-time motion tracking and fall detection with buzzer alert system
- Engineered hybrid dataset combining public and custom annotated data, reducing false positive rate
- Integrated Arducam HM01Bo Monochrome QVGA SPI Camera Module with Raspberry Pi Pico for resource-aware frame capture
- Planned next validation: evaluate false positives and alert responsiveness on recorded motion/fall scenarios, then tune fall rules and thresholds before expanded on-device testing

Technical and Computer Skills

Engineering: CAD/CAM: Fusion 360 (Autodesk), Onshape; NI Multisim, KiCAD, EasyEDA, Soldering

Software: Python, Java, JavaScript, HTML, Jekyll, CSS, Google Workspace, R, Google Sheets, MS Office

Technologies: OpenCV, SolvePnP, TensorFlow, Federated Learning, Kalman Filtering, Finite State Machines, Mechanism Simulation, Control Theory

Work Experience

Math and Reading Instructor, Gideon Math and Reading Center – Frisco, TX June 2024 – present

- Taught groups of ~4–12 students/day from kindergarten to 11th grade in math, reading, and grammar (including Algebra and Geometry), using guided learning sessions
- Analyze student performance data with guided checkpoints to identify learning gaps and develop personalized plans
- Evaluate reading material difficulty and adapt instruction to match target comprehension levels using targeted guided practice

Math and Reading Teaching Assistant (TA), Gideon Math and Reading Center – Frisco, TX Jan 2022 – May 2024

- Graded students' work from kindergarten to 11th-grade, including Algebra and Geometry
- Aided students in completing their assignments appropriate to their age and skill level
- Increased grading efficiency by 50%

Extracurricular Activities

Chief Technical Advisor, FIRST Robotics Competition (FRC) ITKAN Robotics – Frisco, TX Aug 2025 – present

- Provided high-level technical oversight across all robot subsystems for a state-level team of 40 members
- Built an extensive publish-subscribe telemetry and motor logging framework with match logs viewable after events for diagnosing issues and debugging code
- Maintained and extended a >1000-line Java codebase implementing autonomous routines, teleop control, vision, and mechanisms with a command-based paradigm
- Led development of robot state estimation and localization using sensor fusion of AprilTag vision and swerve-drive odometry, achieving <2 inch positional error
- Oversaw development of finite-state machine control architecture with Dijkstra-based path planning using a time-weighted graph
- Used WPILib Java mechanism simulation to reduce time spent on in-house testing by validating behaviors ahead of build
- Engineered complex autonomous routines that optimized the tradeoff between speed and camera (AprilTag) updates, aligning to a specific target over time with PID control
- Achieved a 100% autonomous success rate at the first event
- Produced the 5th-highest-scoring autonomous routine worldwide
- Won an autonomous award at the first event
- Verified performance through testing at home and during the event, supported by online simulations for autonomous pathing
- Developed self-paced robotics programming curriculum and training website (mantik.netlify.app), reaching 1k+ impressions online with users from 6 countries

- Software Lead**, FIRST Robotics Competition (FRC) ITKAN Robotics – Frisco, TX Dec 2024 – July 2025
- Provided technical leadership on software team projects and taught 5-10 students Java programming with a command-based subsystem architecture
 - Taught control theory concepts using PID and motion profiles, and used mechanism simulation with open-source libraries
 - Implemented autonomous Bezier pathing using a command-based architecture to coordinate motion profiles and mechanism actions
 - Advanced team to Texas State Championship playoffs (89 teams)
 - Provided resources to 1000+ people online via my website and a YouTube playlist
 - Robot codebase and documentation: [akhaled247.github.io/first/reefscape](https://github.com/akhaled247/first/reefscape)
- Founder and President**, Wolverine Robotics, Wakeland High School – Frisco, TX Aug 2024 – present
- Founded Wakeland Robotics Club, growing membership from 0 to 50+ students in the first year
 - Developed mission statement, recruited teacher sponsor and members, and presented team vision, schedule, and activities to school administration
 - Secured funding and constructed competition-ready robot in under 2 weeks, advancing to UIL State Championship
 - Teach CAD software (Onshape) and Java programming through monthly technical workshops
 - Manage club operations including curriculum design, calendar management, and budget management
 - Lead monthly club meetings with presentations and demonstrations of robots
- Programming Mentor**, ITKAN Robotics of ICF (FTC 26357, 28931) Sept 2024 – Feb 2026
- Implemented Kalman-filtered state estimation combining AprilTag vision and wheel odometry for real-time robot localization and automated turret targeting
 - Mentored all-rookie FTC team through full engineering design cycle from prototyping to competition
 - Mentor refugee students in FLL workshops and lead weekly summer training for FTC and FRC students using self-developed OARobotics curriculum ([akhaled247.github.io/personal/oarobotics](https://github.com/akhaled247/personal/oarobotics))
- Vice President**, Mu Alpha Theta Math Honor Society, Wakeland High School – Frisco, TX Aug 2024 – present
- Organized and designed presentations for meetings
 - Expanded Math Lab Program and coordinated peer tutoring, attaining over 20% member participation
 - Organized annual middle school math competition with 100+ participants, 3 participating schools, and 35+ volunteers to promote STEM to middle school students
- Treasurer**, Mu Alpha Theta Math Honor Society, Wakeland High School – Frisco, TX Aug 2023 – July 2024
- Designed and implemented Math Lab Program, coordinating student-led tutoring in Algebra I courses with teachers
 - Built website for Math Lab Program using Google Sites for volunteer sign-up
 - Managed budget for 100+ members
- Co-Founder, Secretary**, Muslim Student Association (MSA), Wakeland High School – Frisco, TX Aug 2024 – present
- Developed mission statement, recruited teacher sponsor and members, and presented activities to school administration
 - Coordinated with school staff to establish a prayer room for weekly prayers
 - Created various media for promoting the club and prayer room, including social media posts
- Lead Math and English Instructor**, Ma'ruf Dallas Refugee Organization – Dallas, TX Sept 2023 – present
- Teach Math and English to refugee elementary and middle school children
 - Developed differentiated curriculum for student groups based on their specific needs
 - Provide mentorship to students in academic skills, time management, life skills, and social integration for students transitioning to American schools

Teaching Assistant (TA), Islamic Center of Frisco (ICF) – Frisco, TX

Sept 2023 – May 2024

- Aided elementary students in religious studies and helped keep them on task and engaged
- Assisted Primary Instructor with curriculum implementation

Volunteer, Frisco Family Services – Frisco, TX

Sept 2023 – present

- Sorted and shelved donations for Frisco Family Market, which raises funds for underserved local families with financial, medical, and physical needs
- Sorted and stocked Food Pantry, which also serves underserved local families

Junior Varsity (JV), Wolverines Tennis Team, Wakeland High School – Frisco, TX

Aug 2023 – Aug 2024

- Participated in singles and doubles tournaments throughout academic school year
- Sportsmanship Award Winner 2024

Honors & Awards

- (2026) OralVision: Conrad Challenge Innovator Award & Alternate Finalist, Diamond Challenge Semifinalist
- (2026) FTC 33791: Event Winner, Control Award Winner (FiT-North Dallas Semi-Regional Championship, 25 teams), Control Award Winner (FiT-North U-League Tournament)
- (2025) FRC 9128: FIRST® Leadership Award Semifinalist, Event Winner (FIRST® in Texas Victoria, 25 teams); Autonomous Award (FiT Plano, 31 teams); Creativity Award (FiT Amarillo, 36 teams)
- (2025) FTC 26357: Finalist Alliance Captain (FiT-North McKinney Qualifier, 19 teams)
- (2025) Collegeboard: AP Scholar with Distinction, National Recognition Program: School Recognition Award